

NEW PRODUCTS

TECHNOLOGY

COMPONENTS

Microcontrollers

Microcontrollers that handle mixed-voltage and real-time control are gaining traction for systems where power levels and timing matter. New MCU families from Microchip Technology now support wide supply ranges and on-



chip peripherals that take over critical timing tasks, reducing CPU load. These features suit industrial automation, building control and IoT devices, where precise timing and mixed voltage interfacing improve responsiveness and simplify designs.

Microchip Technology
<https://www.microchip.com>

AI accelerator

Microsoft's Maia 200 AI accelerator is designed to tackle large-scale AI inference by boosting compute, memory bandwidth, and networking for cloud deployments. Running on a 3nm process, it delivers double-digit petaFLOPS performance



and can scale over hundreds of interconnected units, addressing latency and power bottlenecks in real applications. Early adoption in Azure will support demanding services such as Copilot and data-intensive AI platforms, helping deploy models more efficiently.

Microsoft
<https://www.microsoft.com/en-in>

AC-DC power supplies

RECOM's new high-density AC-DC power supplies cram up to 240W into compact DIN-rail formats aimed at industrial control systems where space, efficiency, and reliability matter. Designed for automation, energy

distribution and infrastructure, they handle wide temperature ranges and surge loads without fans, support common industrial voltages, and tolerate back-feeding from inductive loads. These traits help reduce cabinet size, simplify cooling, and maintain stable power in demanding environments.

RECOM Power
<https://recom-power.com/en>

LDO regulators

ROHM's BD9xxN5 LDO regulators deliver up to 500mA output while running stably with ultra-small capacitors ($< 1\mu\text{F}$), overcoming a common design bottleneck.

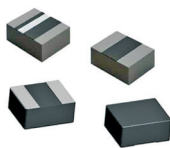


This lets designers shrink power stages without large MLCCs, easing board-space and layout constraints. The regulators suit automotive ECUs, industrial controllers, and communication infrastructure, offering low ripple under fast load swings and greater flexibility in dense 12V/24V power systems.

ROHM Semiconductor
<https://www.rohm.com>

Power inductors

Vishay has rolled out compact, high-performance power inductors in 0806 and 1210 packages that cut board area by up to 64% while maintaining low DC resistance and



stable operation under heat and stress. These smaller inductors help free PCB space and simplify routing in DC/DC converters, filters, and noise suppression applications. Targeted uses

span battery-powered systems, dense computing platforms, and high-current automotive electronics, where size, efficiency, and reliability matter.

Vishay Intertechnology, Inc.
<https://www.vishay.com>

Magnetic switches

Littelfuse's tiny omnipolar magnetic switches pack advanced TMR sensing and CMOS into a compact LGA4 package, operating from 1.8V to 5.5V with ultra-low power draw and high magnetic sensitivity. They fit space-constrained

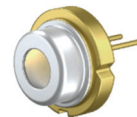


designs and simplify integration by detecting both poles without the need for large magnets. Engineers can use them in healthcare wearables, smart meters, consumer electronics, and automation controls where battery life, size, and fast, precise switching matter.

Littelfuse
<https://www.littelfuse.com>

UV semiconductor laser

A new high-power UV semiconductor laser hitting mass production delivers $\sim 1\text{W}$ continuous output at $\sim 379\text{nm}$ in a compact TO-9 CAN package, pushing mask-



less lithography closer to broader deployment in advanced semiconductor packaging by enabling finer patterning and higher throughput. Its efficient design and thermal management also enable resin curing, precision marking, additive manufacturing, and biomedical instrumentation, where compact, efficient UV sources can replace bulky mercury lamps.

Nuvoton Technology
<https://www.nuvoton.com>

PIR motion sensor

Panasonic Industry's new flat-lens PIR